

Artificial Intelligence

CSIT — 4th Semester — 2079 — Answer Sheet
Subject Code: CSC266

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Define admissible heuristic with an example. Explain the working mechanism and limitations of hill climbing search.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. How do you define problem? What are criteria for defining problem? Compare constraint Satisfaction Problem and Real-World Problem in detail with appropriate example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Define Expert System with example. What are stages of expert system development? Explain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions[8×5=40]

4. How syntactic and semantic analyses are performed in natural language processing?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What do you mean by Rational Agent? What are differences between Utility based agent and model-based agent?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What is state space representation? Illustrate with one example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What is forward chaining? Explain with appropriate example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Convert following sentences into predicatea. All animals who can bark are dog.b. Someone is firing a gunc. All tigers are not fierce

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Define game. Write the benefits and limitations of depth limited search.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. What is fuzzy logic? Discuss the different operators used in genetic algorithm.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Give an example of reinforcement learning. Explain the types of ANN.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write short note of the following (Any TWO):a. Pragmatic Analysisb. Unification and liftingc. Turing test

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.